

Foundations of Bagging ensemble

[Bagging ensemble]

1.0 Content Block

Improving Random Forests and Bagging While the techniques described above utilize random forests and bagging (otherwise known as bootstrapping), there are certain techniques that can be used in order to improve their execution and voting time, their prediction accuracy, and their overall performance. The following are key steps in creating an efficient random forest:

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This process is repeated recursively for successive levels of the tree until the desired depth is reached. At the very bottom of the tree, samples that test positive for the final feature are generally classified as positive, while those that lack the feature are classified as negative. These trees are then used as predictors to classify new data.

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Key Terms There are three types of datasets in bootstrap aggregating. These are the original, bootstrap, and out-of-bag datasets. Each section below will explain how each dataset is made except for the original dataset. The original dataset is whatever information is given.

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History The concept of bootstrap aggregating is derived from the concept of bootstrapping which was developed by Bradley Efron. Bootstrap aggregating was proposed by Leo Breiman who also coined the abbreviated term "bagging" (bootstrap aggregating). Breiman developed the concept of bagging in 1994 to improve classification by combining classifications of randomly generated training sets. He argued, "If perturbing the learning set can cause significant changes in the predictor constructed, then bagging can improve accuracy".

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Creating the bootstrap dataset The bootstrap dataset is made by randomly picking objects from the original dataset. Also, it must be the same size as the original dataset. However, the difference is that the bootstrap dataset can have duplicate objects. Here is a simple example to demonstrate how it works along with the illustration below:

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Finally classifier C^* is generated by using the previously created set of classifiers C_i on the original dataset D , the classification predicted most often by the sub-classifiers C_i is the final classification for $i = 1$ to m . D' = bootstrap sample from D (sample with replacement) $C_i = I(D')$ $C^*(x) = \operatorname{argmax}_{y \in Y} \#\{i: C_i(x)=y\}$ (most often predicted label y)

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Description of the technique Given a standard training set D of size n , bagging generates m new training sets D_i , each of size n' , by sampling from D uniformly and with replacement. By sampling with replacement, some observations may be repeated in each D_i . If $n' = n$, then for large n the set D_i is expected to have the fraction $(1 -$

$1/e$ ($\sim 63.2\%$) of the unique samples of D , the rest being duplicates. This kind of sample is known as a bootstrap sample. Sampling with replacement ensures each bootstrap is independent from its peers, as it does not depend on previous chosen samples when sampling. Then, m models are fitted using the above bootstrap samples and combined by averaging the output (for regression) or voting (for classification).

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Suppose the original dataset is a group of 12 people. Their names are Emily, Jessie, George, Constantine, Lexi, Theodore, John, James, Rachel, Anthony, Ellie, and Jamal. By randomly picking a group of names, let us say our bootstrap dataset had James, Ellie, Constantine, Lexi, John, Constantine, Theodore, Constantine, Anthony, Lexi, Constantine, and Theodore. In this case, the bootstrap sample contained four duplicates for Constantine, and two duplicates for Lexi, and Theodore.

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Further reading Breiman, Leo (1996). "Bagging predictors". *Machine Learning*. 24 (2): 123–140. CiteSeerX 10.1.1.32.9399. doi:10.1007/BF00058655. S2CID 47328136. Alfaro, E., Gámez, M. and García, N. (2012). "adabag: An R package for classification with AdaBoost.M1, AdaBoost-SAMME and Bagging". *Cite journal*: Cite journal requires |journal= (help)CS1 maint: multiple names: authors list (link) Kotsiantis, Sotiris (2014). "Bagging and boosting variants for handling classifications problems: a survey". *Knowledge Eng. Review*. 29 (1): 78–100. doi:10.1017/S0269888913000313. S2CID 27301684. Boehmke, Bradley; Greenwell, Brandon (2019). "Bagging". *Hands-On Machine Learning with R*. Chapman & Hall. pp. 191–202. ISBN 978-1-138-49568-5.

10.0 Content Block

Example: ozone data To illustrate the basic principles of bagging, below is an analysis on the relationship between ozone and temperature (data from Rousseeuw and Leroy (1986), analysis done in R). The relationship between temperature and ozone appears to be nonlinear in this dataset, based on the scatter plot. To mathematically describe this relationship, LOESS smoothers (with bandwidth 0.5) are used. Rather than building a single smoother for the complete dataset, 100 bootstrap samples were drawn. Each sample is composed of a random subset of the original data and maintains a semblance of the master set's distribution and variability. For each bootstrap sample, a LOESS smoother was fit. Predictions from these 100 smoothers were then made across the range of the data. The black lines represent these initial predictions. The lines lack agreement in their predictions and tend to overfit their data points: evident by the wobbly flow of the lines.

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Creating the out-of-bag dataset The out-of-bag dataset represents the remaining people who were not in the bootstrap dataset. It can be calculated by taking the difference between the original and the bootstrap datasets. In this case, the remaining samples who were not selected are Emily, Jessie, George, Rachel, and Jamal. Keep in mind that since both datasets are sets, when taking the difference the duplicate names are ignored in the bootstrap dataset. The illustration below shows how the math is done:

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For classification, use a training set D , Inducer I and the number of bootstrap samples m as input. Generate a classifier C^* as output

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For a weak learner with high bias, bagging will also carry high bias into its aggregate Loss of interpretability of a model. Can be computationally expensive depending on the dataset.

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Specify the maximum depth of trees: Instead of allowing the random forest to continue until all nodes are pure, it is better to cut it off at a certain point in order to further decrease chances of overfitting. Prune the dataset: Using an extremely large dataset may create results that are less indicative of the data provided than a smaller set that more accurately represents what is being focused on. Continue pruning the data at each node split rather than just in the original bagging process. Decide on accuracy or speed: Depending on the desired results, increasing or decreasing the number of trees within the forest can help. Increasing the number of trees generally provides more accurate results while decreasing the number of trees will provide quicker results.